Mathematics Assignment Solutions

Problem 1: Finding the Singular Values (Σ)

Given $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$

Task A (Calculate A^TA):

$$A^TA = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} (1)(1)+(-1)(-1) & (1)(1)+(0)(1)+(-1)(1) \\ (1)(1)+(1)(0)+(1)(-1) & (1)(1)+(1)(1)+(1)(1) \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

The dimensions of A^TA are 2×2. It is a perfectly symmetric diagonal matrix.

Task B (Eigenvalues of A^TA): Since A^TA is a diagonal matrix, its eigenvalues are simply the entries on the main diagonal.

$$\lambda_1 = 3, \quad \lambda_2 = 2$$

Task C (Singular Values): The singular values are the square roots of the eigenvalues of A^TA.

Task D (Construct Σ): To match the 3×2 Shape of A, the full Σ matrix Pads the bottom with zero.

$$\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}$$

Note: In the reduced SVD Commonly used in ML, we just drop the row of zeros, making Σ a 2×2 matrix.

Problem 2: Finding the Right Singular Vectors (V)

Task A ~~(V1 for x)~~ (V_1 for $\lambda_1 = 3$): Sloving $(A^TA - 3I)V = 0 \Rightarrow \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. This means $-x = 0$, so $x = 0$. The value of y can be anything, so we choose $y = 1$ for a unit vector.

Task B (V_2 for $\lambda_2 = 2$): sloving $(A^TA - 2I)V = 0 \Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

This means $y = 0$ The value of x can be anything so we choose $x = 1$. $V_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

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Task C (construct V and V^T):

$$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Longrightarrow V^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Problem 3: Finding the Left Singular Vector (U)

Task A (calculate u_1)

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Task B (calculate u_2):

$$u_2 = \frac{1}{\sigma_2} A v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}$$

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